

## Multiple Choice (10 marks)

1. Imagine that the Earth was instantly moved to an orbit three times further away from the Sun. How much longer would a year be?
  - (a) exactly 3 times longer
  - (b) about 5.2 times longer
  - (c) Not enough information. It will depend on the inclination of the new orbit
  - (d) The length of the year wouldn't change because the Earth's mass stays the same.
2. Say the pupil of your eye has a diameter of 5 mm and you have a telescope with an aperture of 50 cm. How much more light can the telescope gather than your eye?
  - (a) 10000 times more
  - (b) 100 times more
  - (c) 1000 times more
  - (d) 10 times more
3. Which of these has NOT been one of the main hypotheses considered for the origin of the Moon?
  - (a) The Moon split from the Earth due to tidal forces.
  - (b) The Moon was captured into Earth orbit.
  - (c) The Earth and Moon co-accreted in the solar nebula.
  - (d) Earth was rotating so rapidly that the Moon split from it.
4. Which one of these constellations is not located along the Milky Way in the sky?
  - (a) Perseus
  - (b) Cygnus
  - (c) Scorpius
  - (d) Leo
5. In addition to the conditions required for any solar eclipse what must also be true in order for you to observe a total solar eclipse?
  - (a) The Earth must lie completely within the Moon's penumbra.
  - (b) The Moon's penumbra must touch the area where you are located.
  - (c) The Earth must be near aphelion in its orbit of the Sun.
  - (d) The Moon's umbra must touch the area where you are located.

## True / False (10 marks)

*Answer True or False*

6. True or False: The craters at Mercury's poles contain the only permanently shadowed regions on the planet, suitable for preserving water-ice.
7. True or False: The Andromeda Galaxy is approximately 2.5 million light years away from Earth.
8. True or False: The truck would be harder to accelerate on Mars due to its increased weight from the stronger gravitational pull.
9. True or False: The resolution of a telescope is primarily determined by its ability to magnify distant objects.
10. True or False: The Moon's surface is less heavily cratered than the Earth's surface due to more intense geological activity.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss the reasons for the compositional differences between the inner and outer planets in our solar system, focusing on the role of temperature in the condensation of materials during planet formation.
12. Discuss the concept of bolometric luminosity in astronomy, explaining its significance and how it is integrated over all wavelengths in the context of stellar observations.

*End of Paper*